

Deep Learning Systems Analysis Report:

Fashion-MNIST Classification with CNN

Report Overview

This project implements and compares Convolutional Neural Network (CNN) architectures for image classification on the Fashion-MNIST dataset. The baseline model consists of two convolutional layers followed by fully connected layers. An experimental model adds dropout regularization to reduce overfitting. The comparison shows that dropout reduces the training-test accuracy gap from 4.85% to 1.72%, demonstrating effective regularization while maintaining test performance.

Dataset and Task Description

The Fashion-MNIST dataset (Xiao et al., 2017) contains 70,000 grayscale images of fashion items across 10 classes: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle boot. The dataset is split into 60,000 training and 10,000 test images, each 28x28 pixels. This dataset was chosen because it provides a realistic image classification task with balanced classes (6,000 samples per class) and is widely used for benchmarking deep learning models. The task is multi-class image classification, where the model must correctly identify which fashion category each image belongs to.

Model Architecture and Design Decisions

The baseline CNN architecture uses two convolutional layers with ReLU activation and max pooling, followed by two fully connected layers. The first convolutional layer uses 32 filters (3x3 kernel) with padding to preserve spatial dimensions, followed by 2x2 max pooling. The second convolutional layer uses 64 filters (3x3 kernel), also followed by 2x2 max pooling. After flattening, a fully connected layer with 128 units and ReLU activation connects to the output layer with 10 units.

This architecture was chosen because convolutional layers effectively capture spatial hierarchies in images, learning increasingly complex features from edges to shapes to object parts (LeCun et al., 1998). The Adam optimizer was selected for its adaptive learning rate properties, which typically lead to faster convergence and

more stable training compared to standard stochastic gradient descent (Kingma & Ba, 2015). Cross-entropy loss is appropriate for multi-class classification tasks as it measures the divergence between predicted and true probability distributions (Goodfellow et al., 2016). The model was trained for 10 epochs with a batch size of 64.

Experimental Comparison

The controlled experiment introduced dropout regularization after the first fully connected layer while keeping all other components identical. Dropout randomly deactivates a fraction of neurons during training, forcing the network to learn redundant representations and preventing co-adaptation of features (Srivastava et al., 2014). A dropout rate of 0.5 was selected, meaning 50% of neurons are randomly dropped during each training iteration.

The experimental setup differed from the baseline in exactly one major aspect:

- What changed: Added dropout ($p=0.5$) after the first fully connected layer
- What stayed the same: Architecture depth (2 conv layers, 2 fc layers), optimizer (Adam, $\text{lr}=0.001$), batch size (64), epochs (10), loss function (cross-entropy)

This design allows isolation of the regularization effect on model generalization.

Results and Interpretation

The baseline model achieved 96.86% training accuracy and 92.01% test accuracy, with a 4.85% gap indicating moderate overfitting. The experimental model with dropout achieved 93.69% training accuracy and 91.97% test accuracy, reducing the gap to 1.72% - a 3.13% reduction in overfitting.

Per-class analysis revealed that dropout improved accuracy for several classes:

- T-shirt/top: +3.60% (83.8% $\rightarrow$ 87.4%)
- Coat: +2.30% (88.5% $\rightarrow$ 90.8%)
- Ankle boot: +1.30% (95.0% $\rightarrow$ 96.3%)

However, the "Shirt" class decreased from 76.5% to 69.0%, suggesting that Shirt remains the most challenging class due to visual similarity with T-shirt/top and Pullover. The confusion matrix confirms these misclassification patterns, with Shirt frequently confused with T-shirt/top (118 samples) and Pullover (84 samples).

Well-classified categories include Trouser, Sandal, Sneaker, and Bag, all achieving >98% accuracy.

These results demonstrate the trade-off inherent in regularization: dropout slows training convergence (93.69% vs 96.86% training accuracy) but improves generalization, as evidenced by the reduced gap between training and test performance.

Limitations and Risks

Several limitations affect this study. First, the Fashion-MNIST dataset contains only grayscale images of size 28x28, which may not represent real-world fashion classification scenarios where color and higher resolution provide important visual cues. Second, training was limited to 10 epochs due to computational constraints; longer training with early stopping might yield different results. Third, the dropout rate (0.5) was arbitrarily chosen and may not be optimal for this specific architecture. Hyperparameter tuning could identify a more effective rate.

Fourth, the evaluation used only accuracy, which may not capture performance nuances such as class-specific recall or precision (He & Ma, 2013). For example, the model's poor performance on the Shirt class is masked when looking only at overall accuracy. Additional metrics like precision, recall, and F1-score would provide a more complete picture of model performance.

Ethical and Responsible Use

Potential ethical concerns include bias toward certain clothing types if deployed in real-world applications. The dataset was collected from fashion items that may not represent global clothing diversity across cultures and regions (Buolamwini & Gebru, 2018). A model trained primarily on Western fashion items may perform poorly on traditional garments from other cultures, leading to unfair or discriminatory outcomes.

Additionally, such models could be misused for surveillance or profiling based on clothing choices. Responsible deployment would require:

- Diverse training data representing global fashion diversity
- Transparency about model limitations and failure cases
- Clear guidelines restricting use to benign applications like retail inventory management or assistive technologies for visually impaired individuals

- Regular auditing for bias and performance disparities across demographic groups

Future Improvements

Several improvements could enhance this system with additional resources:

1. Hyperparameter tuning: Optimize dropout rate, learning rate, and batch size through systematic experimentation
2. Deeper architecture: Add more convolutional layers or use batch normalization to improve feature learning
3. Data augmentation: Apply random rotations, flips, and shifts to improve generalization and reduce overfitting further (Shorten & Khoshgoftaar, 2019)
4. Transfer learning: Use pre-trained models like ResNet for better feature extraction, especially for challenging classes like Shirt
5. Class-specific improvements: Focus on improving Shirt classification through targeted augmentation, weighted loss functions, or collecting more diverse examples of this class
6. Extended training: Train for more epochs with early stopping to find optimal convergence point
7. Advanced evaluation: Incorporate precision, recall, and F1-score metrics to better understand class-specific performance

References

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